



RIPHAH
INTERNATIONAL UNIVERSITY

BS-AI
LAB TASK 11

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Sap I'D: 72405

Subject: Object Orientated
Programming

Teacher: Sir Imran Khan

Task 1:

```
package com.mycompany.lab12;

abstract class Employee {
    public abstract double calculateSalary();
    public abstract void displayInfo();
}

class Manager extends Employee {
    double baseSalary;
    double bonus;

    Manager(double baseSalary, double bonus) {
        this.baseSalary = baseSalary;
        this.bonus = bonus;
    }

    @Override
    public double calculateSalary() {
        return baseSalary + bonus;
    }

    @Override
    public void displayInfo() {
        System.out.println("Role: Manager");
        System.out.println("Salary: " + calculateSalary());
    }
}

class Programmer extends Employee {
    double baseSalary;
```

```
double baseSalary;
int overtimeHours;
double overtimeRate;
Programmer(double baseSalary, int overtimeHours, double overtimeRate) {
    this.baseSalary = baseSalary;
    this.overtimeHours = overtimeHours;
    this.overtimeRate = overtimeRate;
}
@Override
public double calculatesSalary() {
    return baseSalary + (overtimeHours * overtimeRate);
}
@Override
public void displayInfo() {
    System.out.println("Role: Programmer");
    System.out.println("Salary: " + calculatesSalary());
}
}

public class EmployeeTest {
    public static void main(String[] args) {
        System.out.println("Name: Hafiz Abdullah Shakeel");
        System.out.println("SAP ID: 72405");
        System.out.println();
        Manager manager = new Manager(50000, 10000);
        manager.displayInfo();
        System.out.println();
        Programmer programmer = new Programmer(40000, 20, 500);
        programmer.displayInfo();
    }
}
```

Task 2:

```
package com.mycompany.lab122;
abstract class BankAccount {
    double balance;
    BankAccount(double balance) {
        this.balance = balance;
    }
    public abstract void deposit(double amount);
    public abstract void withdraw(double amount);

    public void displayBalance() {
        System.out.println("Current Balance: " + balance);
    }
}
class SavingsAccount extends BankAccount {
    double interestRate;
    SavingsAccount(double balance, double interestRate) {
        super(balance);
        this.interestRate = interestRate;
    }
    @Override
    public void deposit(double amount) {
        balance += amount;
        System.out.println("Deposited " + amount + " into Savings Account.");
    }
    @Override
    public void withdraw(double amount) {
        if (amount <= balance) {
            balance -= amount;
            System.out.println("Withdrew " + amount + " from Savings Account.");
        }
    }
}
```

```

        balance -= amount;
        System.out.println("Withdrew " + amount + " from Savings Account.");
    } else {
        System.out.println("Insufficient funds in Savings Account.");
    }
}

class CurrentAccount extends BankAccount {
    double overdraftLimit;

    CurrentAccount(double balance, double overdraftLimit) {
        super(balance);
        this.overdraftLimit = overdraftLimit;
    }

    @Override
    public void deposit(double amount) {
        balance += amount;
        System.out.println("Deposited " + amount + " into Current Account.");
    }

    @Override
    public void withdraw(double amount) {
        if (amount <= balance + overdraftLimit) {
            balance -= amount;
            System.out.println("Withdrew " + amount + " from Current Account.");
        } else {
            System.out.println("Withdrawal exceeds overdraft limit.");
        }
    }
}

}

public class BankAccountTest {
    public static void main(String[] args) {
        System.out.println("Name: Hafiz Abdullah Shakeel");
        System.out.println("SAP ID: 72405");
        System.out.println();
        SavingsAccount savings = new SavingsAccount(10000, 0.05);
        savings.deposit(2000);
        savings.withdraw(5000);
        savings.displayBalance();
        System.out.println();
        CurrentAccount current = new CurrentAccount(5000, 2000);
        current.deposit(1500);
        current.withdraw(7000);
        current.displayBalance();
    }
}

```